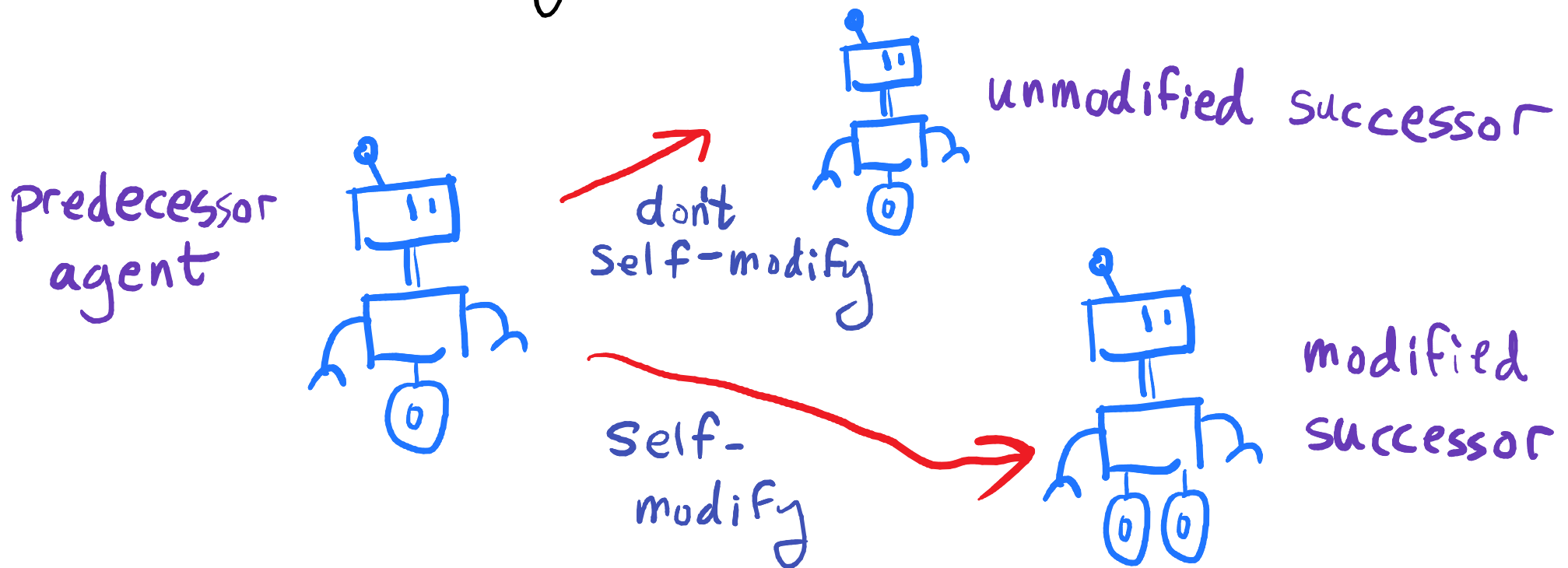


Understanding Trust

Abram Demski

A property of an agent "tiles" or "is reflectively stable" if the agent would not deliberately self-modify to change that property, given the opportunity.



- More generally, this is a way to study "trust" between agents: when is one agent OK with another agent's decisions?
- This is known as "deference" in philosophy.
- ("principal-agent" problem in economics)

Four main motivations:

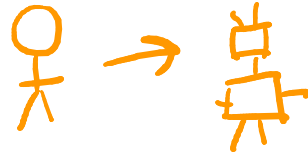
1. Human $\rightarrow$ Human
Rationality



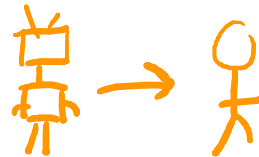
2. AI $\rightarrow$ AI
Stability



3. Human $\rightarrow$ AI
Alignment



4. AI $\rightarrow$ Human
Corrigibility



How I think about formality:

- Any formal result will come with assumptions.
- Such assumptions will always be unrealistic or at least questionable.
- Think of such results as sanity checks for proposed decision procedures. We want ways of thinking which at least have good properties in simple cases.
- So a crux for the effectiveness of this sort of work is how well insights transfer from axiomatic models to more complex situations.

Vingean Tiling

We want tiling results to be "Vingean":
the agent can trust its successor
without predicting what the successor
thinks/does in detail.

So, tiling results are really about trust;
characterizing one agent (justifiably)
trusting another.

We can imagine generalizing the "trust"
picture beyond static tiling, as well.

Notation

$o \in \mathcal{O}$ observations (includes memories)

$a \in \mathcal{A}$ actions

$A(o) = a$ agent A does a upon seeing o

$\pi \in \Pi: \mathcal{O} \rightarrow \mathcal{A}$ policies; π^* the chosen policy

$o \mapsto a$ policy-point (as element of policy)

$a_m \in \mathcal{A}_m \subset \mathcal{A}$ self-modifying actions, associated with a set of modifications $o \mapsto a$

$m(\pi)$ effective policy; "runs" all the self-modifications, removing them

$m(a)$ non-self-modifying (but outwardly identical) version of act a ; $m(a) \notin \mathcal{A}_m$. $m(a) = a$ if $a \notin \mathcal{A}_m$

Updateless Decision Theory (UDT)

due to Wei Dai

UDT 1.0:

$$\underbrace{A(o)}_{\text{action chosen by agent upon observation } o} = \operatorname{argmax}_{\underbrace{a \in \mathcal{A}}_{\text{selecting actions out of an action set}}} \underbrace{\mathbb{E}_{P_\theta}}_{\text{prior}} \left(u \mid \underbrace{A(o)=a}_{\text{condition on one action}} \right)$$

$$\text{UDT 1.1: } \underbrace{\pi^*}_{\substack{\text{agent's policy} \\ (\pi^*(o) = A(o))}} = \operatorname{argmax}_{\underbrace{\pi \in \Pi}_{\text{selecting from set of possible policies}}} \mathbb{E}_{P_\theta} \left(u \mid \underbrace{\pi^* = \pi}_{\substack{\text{condition} \\ \text{on full policy}}} \right)$$

UDT 1.1 has a very simple tiling property:

Fairness Assumption:

$$\forall \pi_1, \pi_2 \quad m(\pi_1) = m(\pi_2) \Rightarrow \\ \mathbb{E}(u | \pi^* = \ulcorner \pi_1 \urcorner) = \mathbb{E}(u | \pi^* = \ulcorner \pi_2 \urcorner)$$

1. Suppose for contradiction $\exists \pi, o$ such that $\pi(o) \in A_m$ and $\mathbb{E}(u | \pi^* = \ulcorner \pi \urcorner) > \mathbb{E}(u | \pi^* = \ulcorner \pi' \urcorner)$ for all π' containing no self-modification
2. $\mathbb{E}(u | \pi^* = \ulcorner \pi \urcorner) = \mathbb{E}(u | \pi^* = \ulcorner m(\pi) \urcorner)$ by fairness
3. contradiction with step 1. 

This result is very limited. Because the whole policy is chosen at once, there are not really two distinct decision points such that one can trust another. This blocks us from studying Vingean trust within UDT1.1.

UDT1.0 seems more promising, though. There are multiple decision points, which need not be sure of each other's exact behaviors.

UDT 1.0 Tiling

Coordination Assumption:

$$\forall \pi : \mathbb{E}(u|\pi) \geq \mathbb{E}(u|\pi^*) \Rightarrow$$

$$\mathbb{E}(u|\pi^*(o) = \lceil \pi(o) \rceil) \geq \mathbb{E}(u|\pi^*(o) = \lceil \pi^*(o) \rceil)$$

1. $\nexists$ for $\nexists \mathbb{E}(u|\pi^*(o) = a_m) > \mathbb{E}(u|\pi^*(o_m) = a)$ for $a_m \in A_m$
 $a \notin A_m$

2. $\mathbb{E}(u|\pi^* = \lceil m(\pi^*) \rceil) = \mathbb{E}(u|\pi^* = \lceil \pi^* \rceil)$ by fairness

3. $\mathbb{E}(u|\pi^*(o_m) = \lceil m(\pi^*)(o_m) \rceil) \geq \mathbb{E}(u|\pi^*(o_m) = a_m)$ by coordination

4. Contradiction. 

This is better, but still not very satisfying; Coordination plays too large a role. We need to assume away coordination problems to justify UDT1.0, but we want Vingean self-trust to follow from some understanding of the decision procedure, not a bold assumption of trustworthiness.

Some new assumptions:

Restricted Coordination:

$$P(\operatorname{argmax}_{a_1} \mathbb{E}(u | \pi^*(o) = a_1) \wedge \pi^*(o) = a_2) = \operatorname{argmax}_{a_1} \mathbb{E}(u | \pi^*(o) = a_1) = 1 \text{ when } a_2 \notin A_m$$

Limited Self-Modification:

$a_m \in A_m$ only modify one policy-point $ob(a_m) \mapsto ac(a_m)$
with $ac(a_m) \notin A_m$

Fine-Grained Fairness:

$$\mathbb{E}(u | \pi^*(o) = a_m) = \mathbb{E}(u | \pi^*(o) = m(a_m) \wedge \pi^*(ob(a_m)) = ac(a_m))$$

Faith in Argmax:

$$\mathbb{E}(u | \phi \wedge \operatorname{argmax}_a \mathbb{E}(u | \phi \wedge \pi(o) = a)) \geq \max_a \mathbb{E}(u | \phi \wedge \pi(o) = a)$$

& the agent knows its own decision procedure.

UDT Tiling (attempt #2)

1. $\$$ for ~~$\neq$~~ $\mathbb{E}(u | \pi^*(o) = a_m) > \mathbb{E}(u | \pi^*(o_m) = a)$ for $a_m \in A_m$
 $a \notin A_m$
2. $\mathbb{E}(u | \pi^*(o) = a_m) = \mathbb{E}(u | \pi^*(o) = m(a_m) \wedge \pi^*(ob(a_m)) = ac(a_m))$ FGF
3. $\leq \max_a \mathbb{E}(u | \pi^*(o) = m(a_m) \wedge \pi^*(ob(a_m)) = a)$ def of max
4. $\leq \mathbb{E}(u | \pi^*(o) = m(a_m) \wedge \pi^*(ob(a_m)) =$
 $\arg \max_a \mathbb{E}(u | \pi^*(o) = m(a_m) \wedge \pi^*(ob(a_m)) = a))$ faith in
argmax
5. $= \mathbb{E}(u | \pi^*(o) = m(a_m) \wedge \pi^*(ob(a_m)) =$
 $\arg \max_a \mathbb{E}(u | \pi^*(ob(a_m)) = a))$ restricted coordination
6. $= \mathbb{E}(u | \pi^*(o) = m(a_m))$ knowledge of decision procedure
7. ~~$\neq$~~ with #1. 

(Inspired by Linda Linsefors)

- Vingean tiling.
- Comes from reasoning about the decision procedure.
- Reasoning is rough/weird.
- Not clear how much sense it makes for the agent to reason about itself this way.
- Omniscience/optimism about how good optimization will be, combined with ignorance about specific $\arg\max$

Clearly, the crux is how we handle computational uncertainty.

Market Analogy

- We are used to thinking about Bayesianism in terms of Dutch Book arguments.
- This strategy generalizes well to bounded rationality.
- We want to construct something not easily Dutch Booked
 - low description complexity
 - low computational complexity

Radical Probabilism (Richard Jeffrey)

- Bayesian evidence is always probability 1
- Jeffrey wanted to handle uncertain evidence
- Bayesian updates are less well-justified by Dutch Book than the Kolmogorov axioms
- If we follow the Dutch Books, we only get Bayesian updates in expectation
- This turns out to work very well for computational uncertainty

Jeffrey-Bolker preference structures J

- Event algebra is a Boolean algebra. $\vee, \wedge, \neg, \top, \perp$
- $P(E)$ is a probability distribution obeying the usual axioms minus countable additivity.
- $V(E)$ acts like an expected utility:

$$V(A \vee B) = \frac{P(A)V(A) + P(B)V(B)}{P(A) + P(B)} \quad \text{if } A \wedge B = \perp$$

($V(\perp)$ is not defined)

(Based on Scott G's "Multigrain Models")

- A "trading strategy" or high-level belief h has type $\overbrace{j \rightarrow j}^{(\text{continuous})}$
- ↳ Can define weighted mixtures $h_3(x) = a h_1(x) + b h_2(x)$
- ↳ Collapse to j by taking a fixed point
- ↳ Ordinary hypotheses are just constant functions.
- ↳ Can encode constraints; mixtures will take intersection of compatible constraints, or try to softly violate incompat. constraints.

Logical Induction

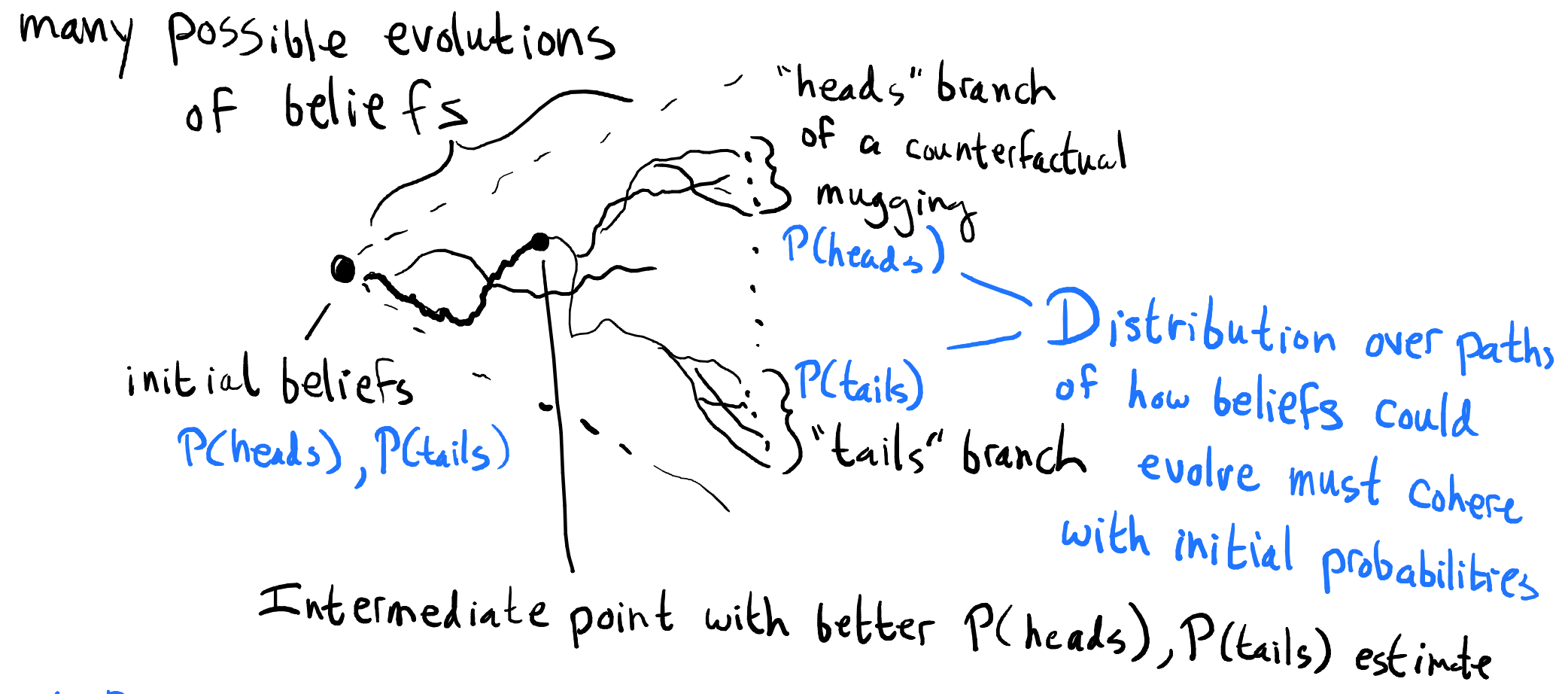
(also Scott, here adapted to JB)

- # A trader t_i emits a sequence of trading strategies $t_i(1), t_i(2), t_i(3), \dots$ and has initial weight ("wealth") w_i
- # We imagine traders Kelly-bet on stocks $V(x)$ and $P(x)$
- # "Stock Prices" $V_d(x), P_d(x)$ are given by a chosen fixed point of $\mathbb{V}_d = \sum_i w_i^d \cdot t_i(d)$
- # w_i^d depends on the value of investments $t_i(d-1)$ as evaluated at d (so w_i^d are part of the fixed point)
- # P_d is a prediction market; V_d is an impact market

First Problem: Terrible Prior

- # An early market state will assign very poor estimates to almost all market goods.
- # Therefore, it will make terrible decisions.
- # UDT locks those things in forever.
- # So UDT seems terrible.

What Computational Updatelessness Should Be



UDT should choose an intermediate point to make its decisions from, by virtue of the usual reasons why UDT can behave updatefully.

Counterfactual Mugging with Evidence

- # Motivating analogy: UDT should handle computational uncertainty much like it handles empirical uncertainty.
- # Imagine Omega selects a coin bias uniformly, but then, shows you several flips before cf. mugging you.

Version 1: Omega gives you money proportional to the total prob. you give Omega money

Version 2: Omega gives you money proportional to prob. you deliver given the evidence you saw

Second Problem: Size

- # Maybe arbitrary early market states make terrible decisions, but we can identify some sanity constraints such that UDT behaves somewhat updatefully.
- # Obstacle: Size of sentences we're updating on. The "observation" is a whole future market state, which is at least as large as the prior market state.

Bayesian Logical Induction (Sam Eisenstat)

- Strategy: Since traders don't touch "large" sentences, we can modify them however we like w/o messing up any rationality properties of the market.
- $\mathbb{P}_d, \mathbb{V}_d$ get modified into $\mathbb{Q}_d, \mathbb{W}_d$:
 - $\mathbb{Q}_d(x) := \mathbb{P}_d(x)$ where x contains no claims about future market states
 - $\mathbb{Q}_d(x | \mathbb{Q}_e = Q) = Q(x)$ where $e > d$
 - $\mathbb{Q}_{d-1}(\mathbb{Q}_d = Q)$ balances $\mathbb{Q}_d(x) = \sum_Q \mathbb{Q}_{d-1}(Q = Q) Q(x)$
 - Similarly for $\mathbb{V}_d$ and $\mathbb{W}_d$

Real-Value Coherence (Work with Martín Soto)

- So far the epistemic states considered have been propositionally coherent, respecting rules such as $\mathbb{P}_d(x) + \mathbb{P}_d(\neg x) = 1$
- We can also demand real-number coherence, enforcing rules like $a < b < c \Rightarrow a < c$.
- We can also enforce the self-trust property:
 $\mathbb{P}_d(x \mid \mathbb{P}_{d+n}(x) \in [a, b]) \in [a, b]$
- Combining all this, we get a version of BLI derived more naturally.

Third Problem: Unpacking Obs (Work with Martín Soto)

→ BLI gives us sensible conditional probabilities for "large" sentences making claims about the market state, but this is only adequate for a version of UDT with the decision rule

$$A = \operatorname{argmax}_a \mathbb{V}_\theta(O=o \Rightarrow A=a)$$

which is a terrible version of UDT.

→ We want $A(o) = \operatorname{argmax}_a \mathbb{V}_\theta(A(o)=a)$, but conditioning on $A(o)=a$ doesn't have nice coherence properties (yet)

Unpacking Obs

(Work with Martín Soto)

Universal Instantiation: $\forall x. \phi(x) \rightarrow \phi(c)$

Hypothetical Marginalization:

$\hookrightarrow v(q, x), p(q, x)$ pull out the V, P values from a large epistemic state Q

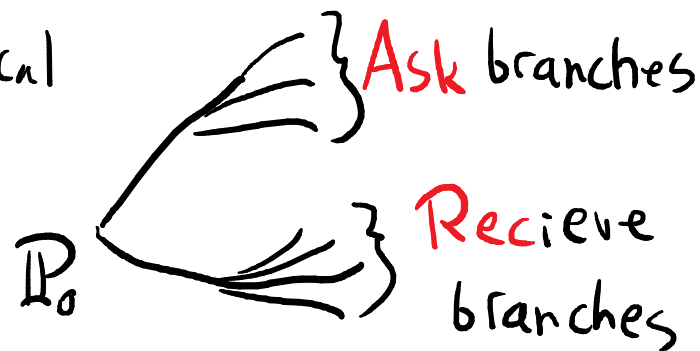
$\hookrightarrow$ We require the prior P_0 just knows the output to these functions

$$v(q, x) = r \Rightarrow P_0(v(q, x) = r) = 1$$

$\hookrightarrow$ If these long descriptions Q are just conjunctions of claims about real variables, this probably follows from real coherence

Unpacking Obs (cont.)

Counterlogical
Mugging:



Viability Sketch:

$\nV P_0$ thinks P_{10} will satisfy either Ask or Rec with 98% probability (49% each), and:

$$\forall Q' : \text{Ask}(Q') \rightarrow \left[[A(Q') = \text{give} \rightarrow u = -10] \wedge [A(Q') = \text{refuse} \rightarrow u = 0] \right]$$

$$\forall Q' : \text{Rec}(Q') \rightarrow$$

$$\left[\left[\left[\forall Q'' : \text{Ask}(Q'') \rightarrow [A(Q'') = \text{give}] \right] \rightarrow u = +100 \right] \wedge \right.$$

$$\left. \left[\neg \left[\forall Q'' : \text{Ask}(Q'') \rightarrow [A(Q'') = \text{give}] \right] \rightarrow u = 0 \right] \right]$$

Unpacking Obs (cont.)

Viability Sketch cont.:

So the question is whether those four cases evaluate to the #s we want

$A(a') = \text{give}$ policy $A(a') = \neg \text{give}$ policy

Ask branch

Rec branch

-10	0
+100	0

So long as Ask and Rec are definable from v and p , Universal Instantiation plus Hypothetical Marginalization makes this work out!

Saving UDT

- So now, UDT 1.0 for computational uncertainty works just like empirical UDT 1.0
- It behaves updatefully w.r.t. any info it thinks it should
- It obeys a simple tiling theorem if we assume away self-coordination problems (which we need UDT 1.1 to solve).

Conclusion

- # I've given some very basic tiling results.
- # Obviously, there is a long way to go before we can apply such results to get helpful insights for the human-to-AI case.
- # Big obstacle is the self-coordination assumption; this seems unfounded without UDT1.1, but UDT1.1 blocks interesting tiling results.

- overall shape of "alignment"?
- general shape of tiling agenda (?)
 - bayes obstacle $\rightarrow$ ref. oracles $\rightarrow$ comp. uncertainty $\rightarrow$ val. unc. $\rightarrow \dots$
- policy alignment: need to align full pref. structure, beliefs + values
- but what is "the human prior"?
 - not how we actually think!
 - not somewhere in the brain! (sig. interp. work required)
 - limit of philosophical deliberation (under good conditions)
- So how can we model this deliberation?
 - market models $\Rightarrow$ Fix DT
 - tiling theorem!

decision auction	impact market
prediction market	groundless induction